

Privacy-Preserving Federated Learning for Tuberculosis Detection from Chest X-Rays Across Nigerian Teaching Hospitals

Week 4 Project Report

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Architecture: ResNet-18 (ImageNet pretrained)

Dataset: Montgomery County + Shenzhen Hospital CXR ($N = 800$)

Date: June 18, 2026

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Appendix A

Mathematical Derivations

A.1. Dirichlet Partitioning: Formal Derivation

Given K clients, a total training set $\mathcal{D}$, and a Dirichlet concentration parameter $\alpha > 0$, the partition assigns each sample $(\mathbf{x}, y)$ to exactly one client as follows.

For each class $c \in \{0, 1\}$, let $\mathcal{D}_c$ be the set of training samples with label c . Draw a probability vector:

$$\mathbf{q}_c = (q_{c,1}, \dots, q_{c,K}) \sim \text{Dir}(\alpha \mathbf{1}_K).$$

Assign $\lfloor q_{c,k} \cdot |\mathcal{D}_c| \rfloor$ samples from $\mathcal{D}_c$ (after random shuffling) to client k . The client k 's dataset is $S_k = \mathcal{D}_0^{(k)} \cup \mathcal{D}_1^{(k)}$.

The expected class proportion at client k is:

$$\mathbb{E} \left[\frac{|S_k^+|}{|S_k|} \right] = \frac{\mathbb{E}[q_{1,k}] \cdot |\mathcal{D}_1|}{\mathbb{E}[q_{0,k}] \cdot |\mathcal{D}_0| + \mathbb{E}[q_{1,k}] \cdot |\mathcal{D}_1|} = \frac{|\mathcal{D}_1|}{|\mathcal{D}|} = \text{global TB+ rate}.$$

In expectation, each client has the global class balance. The variance (which drives heterogeneity) is:

$$\text{Var}[q_{c,k}] = \frac{\alpha}{K(K\alpha + 1)} \xrightarrow{\alpha \rightarrow \infty} 0.$$

A.2. Sequential Composition Theorem: Proof Sketch

Theorem A.1 (Sequential Composition). If $\mathcal{M}_1$ is (ϵ_1, δ_1) -DP and $\mathcal{M}_2$ is (ϵ_2, δ_2) -DP and they are applied sequentially to the same dataset, the composition $(\mathcal{M}_1, \mathcal{M}_2)$ is $(\epsilon_1 + \epsilon_2, \delta_1 + \delta_2)$ -DP.

Proof sketch. For adjacent datasets D, D' , and output pairs $(o_1, o_2) \in S$:

$$\mathbb{P}[(\mathcal{M}_1(D), \mathcal{M}_2(D, o_1)) \in S] \quad (\text{A.1})$$

$$\leq \int_{o_1} \mathbb{P}[\mathcal{M}_2(D, o_1) \in S_{o_1}] \cdot \mathbb{P}[\mathcal{M}_1(D) = o_1] do_1 \quad (\text{A.2})$$

$$\leq \int_{o_1} (e^{\varepsilon_2} \mathbb{P}[\mathcal{M}_2(D', o_1) \in S_{o_1}] + \delta_2) \cdot \mathbb{P}[\mathcal{M}_1(D) = o_1] do_1 \quad (\text{A.3})$$

$$\leq e^{\varepsilon_2} \int_{o_1} \mathbb{P}[\mathcal{M}_2(D', o_1) \in S_{o_1}] \cdot (e^{\varepsilon_1} \mathbb{P}[\mathcal{M}_1(D') = o_1] + \delta_1) do_1 + \delta_2 \quad (\text{A.4})$$

$$\leq e^{\varepsilon_1 + \varepsilon_2} \mathbb{P}[(\mathcal{M}_1(D'), \mathcal{M}_2(D', \cdot)) \in S] + e^{\varepsilon_2} \delta_1 + \delta_2 \quad (\text{A.5})$$

$$\leq e^{\varepsilon_1 + \varepsilon_2} \mathbb{P}[(\mathcal{M}_1(D'), \mathcal{M}_2(D', \cdot)) \in S] + \delta_1 + \delta_2, \quad (\text{A.6})$$

which establishes $(\varepsilon_1 + \varepsilon_2, \delta_1 + \delta_2)$ -DP. $\square$

A.3. Gaussian Noise Multiplier Derivation

For the Gaussian mechanism with clipping norm C , batch size B , dataset size n , and T training steps, the noise multiplier σ required to achieve (ε, δ) -DP satisfies:

$$\varepsilon \approx \frac{q \sqrt{T \ln(1/\delta)}}{\sigma}, \quad q = \frac{B}{n},$$

which gives (approximately):

$$\sigma \approx \frac{q \sqrt{T \ln(1/\delta)}}{\varepsilon}.$$

This is the simplified ‘‘Gaussian mechanism’’ formula. Opacus uses a tighter RDP-based computation that accounts for the sub-sampled Gaussian mechanism, yielding lower σ for the same (ε, δ) guarantee.

A.4. McNemar’s Test Derivation

For two classifiers A and B evaluated on n test samples, define the contingency table:

	B correct	B wrong
A correct	a	b
A wrong	c	d

Under H_0 (no difference in error rates), b and c follow a Binomial($b + c, 0.5$) distribution. For large $b + c$, the test statistic:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \sim \chi_1^2.$$

The correction -1 in the numerator (Yates' continuity correction) is applied because the binomial is discrete but the chi-squared is continuous. For our comparison: $b = 29$, $c = 12$, giving $\chi^2 = 16^2/41 = 6.24$, $p = 0.012$.

A.5. Youden Index and Threshold Selection

The Youden index at threshold τ :

$$J(\tau) = \text{TPR}(\tau) - \text{FPR}(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1.$$

The optimal threshold $\tau^* = \arg \max_{\tau} J(\tau)$ maximises the sum of sensitivity and specificity, which is equivalent to maximising the vertical distance from the ROC curve to the diagonal chance line. This criterion assumes equal costs for FP and FN errors. For asymmetric costs $c_{\text{FN}} \neq c_{\text{FP}}$, the optimal threshold is:

$$\tau^* = \arg \max_{\tau} [c_{\text{FN}} \cdot \text{Sensitivity}(\tau) + c_{\text{FP}} \cdot \text{Specificity}(\tau)],$$

which shifts the threshold towards lower τ (higher sensitivity) when $c_{\text{FN}} \gg c_{\text{FP}}$.

Appendix B

Complete Numerical Results Tables

B.1. Full Metrics Table: All Three Models

Table B.1. Full test set metrics for all three model variants. “–” indicates the metric was not bootstrapped for this run.

Metric	Centralised	FL (No DP)	FL+DP ($\varepsilon = 8$)
AUC-ROC	0.8939	0.9086	0.7296
AUC-ROC CI lower	0.8330	0.8527	0.6416
AUC-ROC CI upper	0.9430	0.9526	0.8147
AUC-PRC	0.9113	0.9164	0.7279
AUC-PRC CI lower	0.8548	0.8582	0.6149
AUC-PRC CI upper	0.9539	0.9601	0.8477
Sensitivity	0.8136	0.7627	0.7627
Sens CI lower	0.7169	0.6441	0.6525
Sens CI upper	0.9034	0.8627	0.8644
Specificity	0.8525	0.9180	0.6230
Spec CI lower	0.7541	0.8443	0.5082
Spec CI upper	0.9344	0.9836	0.7377
PPV	0.8421	0.9000	0.6618
NPV	0.8254	0.8000	0.8046
F1-Score	0.8276	0.8257	0.7087
Balanced Accuracy	0.8330	0.8404	0.6928
Accuracy	0.8333	0.8417	0.6917
TP	48	45	45
TN	52	56	38
FP	9	5	23
FN	11	14	14

B.2. Statistical Comparison Table

Table B.2. Statistical comparison between centralised and FL+DP models.

Statistic	Value
McNemar b (Central correct, FL+DP wrong)	29
McNemar c (FL+DP correct, Central wrong)	12
McNemar χ^2	6.2439
McNemar p -value	0.0124
Δ AUC (Central – FL+DP)	0.1642
Δ AUC Bootstrap CI lower (2.5%)	0.0755
Δ AUC Bootstrap CI upper (97.5%)	0.2538
Bootstrap p -value (two-sided)	0.5082
Non-inferiority margin	0.05
Non-inferiority established?	No

B.3. FL Round-by-Round Results

Table B.3. FL (No DP) round-by-round test metrics.

Round	Test Loss	Test AUC-ROC	Delta AUC
0	0.7766	0.4173	–
1	0.5527	0.8527	+0.4354
2	0.4692	0.9069	+0.0542
3	0.4999	0.8930	–0.0139
4	0.4419	0.9069	+0.0139
5	0.5377	0.9086	+0.0017

B.4. Ablation: FL Rounds

Table B.4. FL (No DP) test AUC vs number of communication rounds.

Rounds	AUC-ROC	Comm. Cost (5 sites, MB)
1	0.8639	215
2	0.8761	430
3	0.8755	645

B.5. Ablation: Dirichlet α

Table B.5. FL (No DP) test AUC vs Dirichlet heterogeneity parameter α .

α	Site 0 TB+ Rate	AUC-ROC	Interpretation
0.1	100.0%	0.3851	FL fails (extreme heterogeneity)
0.5	51.0%	0.8183	FL succeeds (moderate heterogeneity)
1.0	23.4%	0.6588	FL underperforms (adverse draw)

B.6. Per-Site Fairness Analysis

Table B.6. Per-site test performance (centralised model) showing the equity gap across simulated Nigerian teaching hospitals.

Site	n	AUC	Sens	Spec
Lagos Univ. Teaching Hosp.	23	0.9524	1.0000	0.8889
Aminu Kano Teaching Hosp.	17	0.9091	0.8333	0.8182
UNTH Enugu	27	0.8556	0.8000	0.8333
OAU Teaching Hosp.	32	0.8392	0.6667	0.8235
UCH Ibadan	21	0.9352	0.7778	0.9167
Equity Gap (max–min AUC)		0.1132		